

Lecture11-Bedform

November 17, 2025

1 Lecture11 - Coastal sediment transport

Learning Objectives: ripples vs dunes, erosion and deposition, bedform formation (wave vs current), bedform migration (lower vs upper flow), longshore sediment transport, hard vs soft coastal protection methods

Before class:

- Announcement: Group presentation days 12/2 (3-4 groups) and 12/4 (1-2 groups); final's prep day 12/4; final's exam 12/12 @ 8am!

After class:

- Read about the [“Large aeolian ripples on Mars”](#) and notice the size difference in wind-blowing ripples on Earth vs Mars.
- Revisit [100 years change in Connecticut's Coast](#) and consider the following: What factors have driven changes along Connecticut's coastline? Which areas have experienced the most significant changes? How might the coast continue to evolve in the future?

Reference:

- classnote taken from CEE262G Sediment Transport Physics and Modeling by Oliver Fringer, Stanford University
- [Dr. Glessmer's blog on Sand ripples. Why do they form in the exact way they do?](#)
- Julien Ch. 8
- B.S. Ch. 7.3, 8.2, 8.4

1.1 1. Bedforms

Ripples, dunes, and antidunes are all bedforms, structures that form in sediment when it is moved by water or wind. Bedforms are ubiquitous on our planet. It is very common to see **ripples**, undulatory structures in sand, under shallow waters close to seashores or along riverbanks. And deserts, they are commonly covered with large sand **dunes**, in turn sprinkled by smaller sand waves, all structures produced by wind. All these structures have a strong preservation potential and can survive in the rock record (fossil ripple marks or [cross-bedding](#) that indicate migrating bedforms in the past.

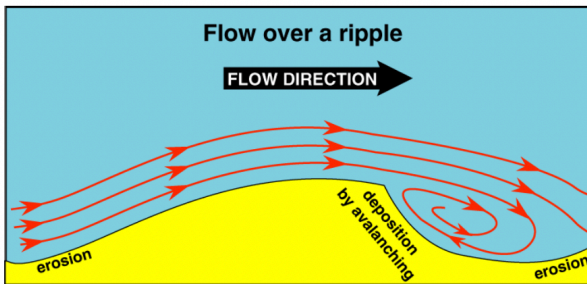
1.1.1 1). Ripples and dunes

Ripples

- Small scale, asymmetric bed forms.
- Develop on sands with $d_s < 0.7$ mm
- Migrate downstream (in the direction of the lee slope).
- $0.05 < \text{Length} < 0.6$ m
- $0.005 < \text{amplitude} < 0.05$ m
- Length scales with grain size: $\lambda \approx 1000 d_s$
 $\rightarrow \lambda^+ \approx 1000 Re_*$



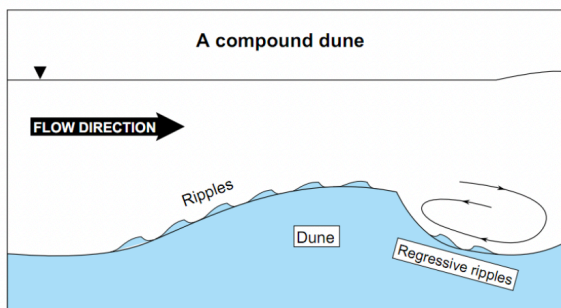
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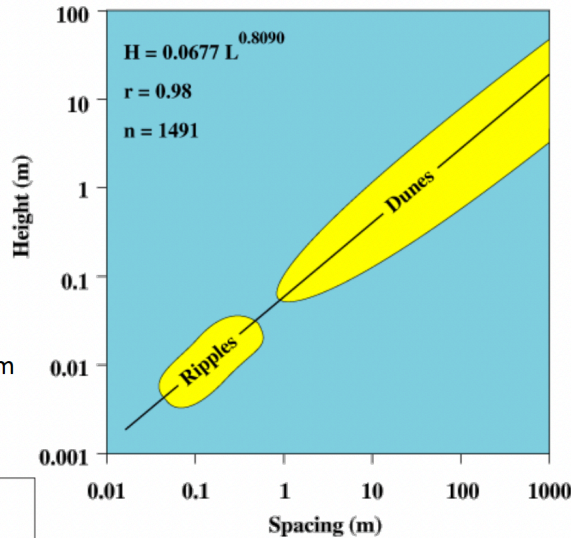
McMahon and Tobicoe, Sedimentology Lecture Notes
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Dunes

- Large, asymmetric bed forms.
- Not “large ripples” but a dynamically different bed form since no longer scales with grain size but instead with depth and flow properties (i.e. eddies)
- Length ranges from 0.75 m to over 100 m
- Amplitude 7.5 cm to over 5 m
- Most common in sands coarser than 0.15 mm
- Also referred to as “megaripples”



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<http://www.amustard.com/?page=prints/pr21&subpage=prints/pr21>

1.1.2 2). Bedform formation

Discussion:

Watch this video on River Sediment Processes and discuss

- 1) Do you observe ripples or dunes in the river?
- 2) What factors control sediment erosion versus deposition?
- 3) What processes might drive the formation and migration of ripples and dunes?

a. Wave-induced ripples Now watch another video on a lab experiment demonstrating the formation and migration of sand ripples. Does the video confirm your earlier observations?

Recall that in Lecture 04 we discussed the differences between shallow-water and deep-water waves.

B.

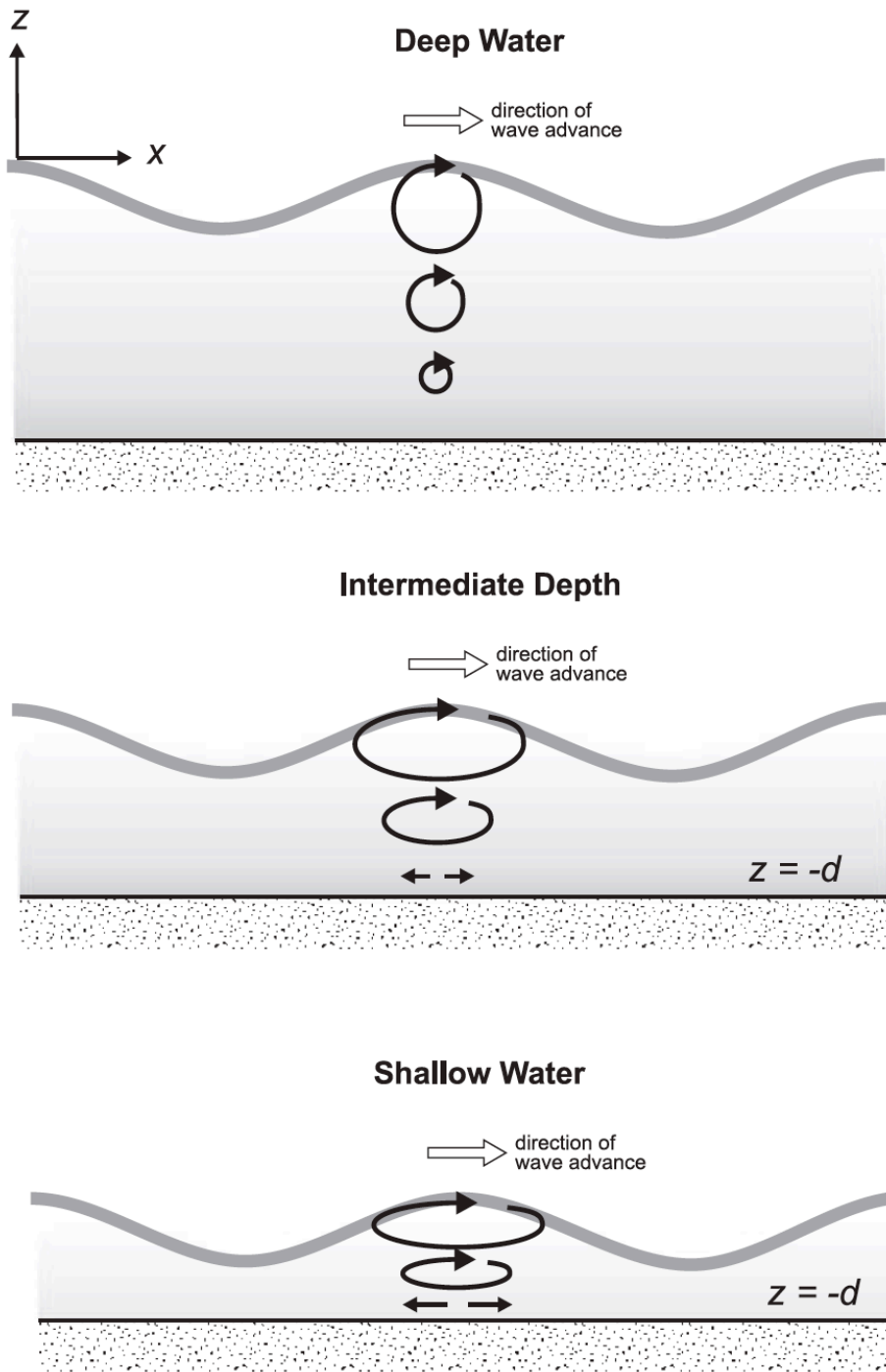


Figure 4.1 Wave characteristics: (A) definition sketch of the characteristics of a simple progressive wind wave in deep water; (B) wave orbital motion in deep, intermediate and shallow water.

One of the most important characteristics of waves in shallow water is that the sea floor feels their influence because the orbital motion of the waves becomes elongated into ellipses, creating a back-and-forth current near the bottom. Ripples formed by waves are typically more **symmetric** and do not migrate. **That’s why you can observe sand ripples on the foreshore where the water is shallow!**

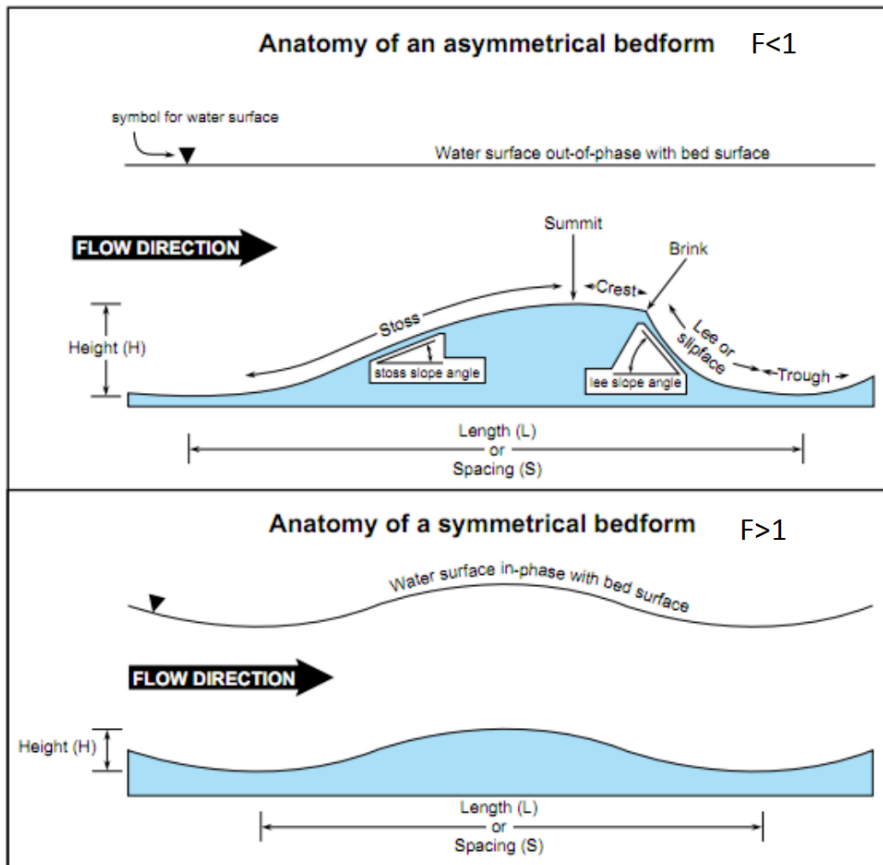
The video above shows a numerical simulation of sand ripple formation by waves (upper panel) and by a combination of waves and currents (lower panel), created by Oliver Fringer. Note that ripples formed by waves alone do not migrate, unlike those formed by waves combined with currents.

b. Ripple scale

- The influence of **current speed**
 - The faster the current, the longer the wavelength of the ripple. This makes sense since in stronger currents, grains are transported further away from the ripple they were eroded from until they deposit again and form the next ripple.
 - Stronger currents are probably linked to longer waves (with larger orbitals) and shallower water.
- The influence of **grain size**
 - If sand grains are too small, they don’t form ripples (or they are just extremely flat). However, if the grains are more like stones, they are not moved into ripples anymore, either, but instead create a turbulent flow around them. But for the happy medium in between, the larger the grain, the larger the ripple (both in length and in amplitude).
 - A general rule for length scales relative to grain size is: $\lambda = 1000d_s$.
 - Dunes, sometimes referred to as “megaripples,” are a dynamically different bedform from ripples. Unlike ripples, their size no longer scales with grain size but instead depends on water depth and flow characteristics (e.g., turbulent eddies).
- Of course, it’s not only grain size, it’s also tons of other factors like grain shape (that will influence cohesion between grains) or bio growth like bacterial mats, that will also make it more difficult for grains to be moved.

1.1.3 3). Bedform migration

a. Current-induced ripples **Asymmetric ripples** form at the contact between water/air and the sediment dragged by a unidirectional current. Imagine a current over a sandy bottom. If the bottom isn’t completely flat but there is a tiny imperfection somewhere, this will influence the bottom current: speeding it up the higher the bump is (i.e. the smaller the cross section of the current becomes) and slowing it down when the cross section widens again. So a current flowing over a bump will be sped up. Looking closely at the bump, it has an asymmetric profile: A gradual incline on the upstream side (apparently called the “**stoss side**”) and a steep drop on the **lee side**. Sand is being **eroded** from the upstream side where the current is strong, and being **deposited** on the lee side where the current is weak, thus slowly moving the bump forward in the direction of the current.

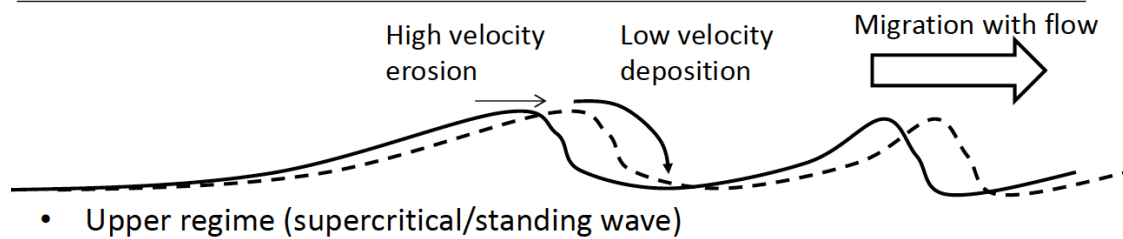


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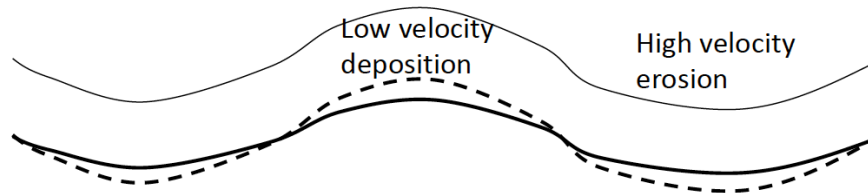
This [video](#) demonstrates the migration of sand ripples in the same direction as the current.

b. Bedform migration by the current As we discussed in the formation of symmetric ripples by waves, the primary parameter controlling bedforms is the current velocity. The current must first reach a **critical velocity**, beyond which it has sufficient energy to initiate sediment transport. Once this threshold is reached, sediments begin moving by traction along the bed, and ripples start to form.

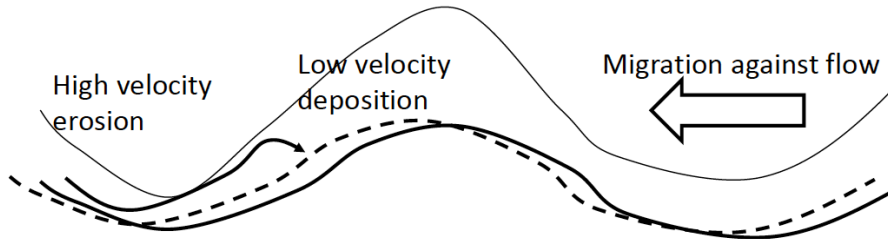
- Lower regime (subcritical)



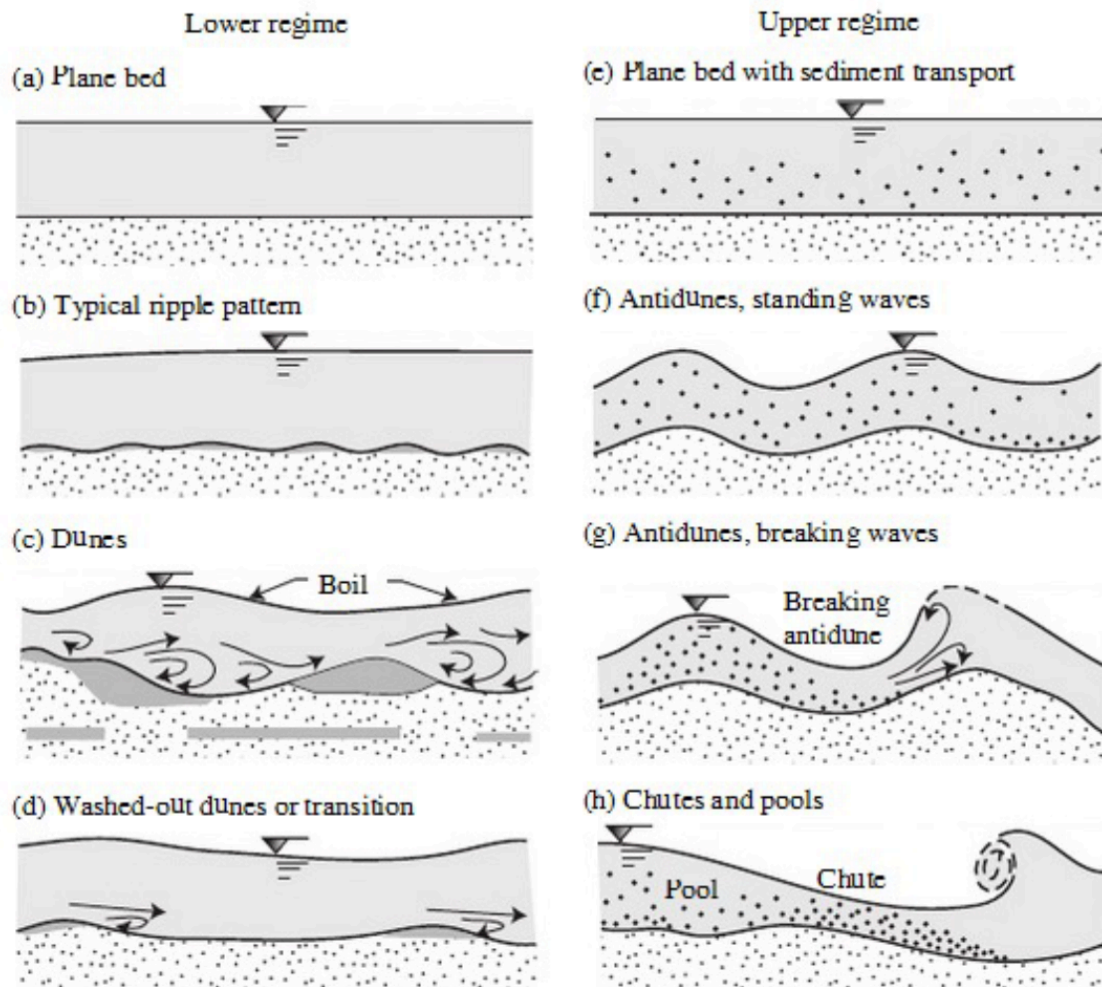
- Upper regime (supercritical/standing wave)



- Upper regime (supercritical/antidune)



Flow regimes influence bedform types and their characteristics. - In the **lower flow regime**, bedforms include lower plane beds, ripples, and dunes. This regime is characterized by a small flow rate, weak sediment transport dominated by bedload, and bedforms that are out of phase with the free surface. - In contrast, the **upper flow regime** features upper plane beds, in-phase waves, and chutes and pools. Here, the flow rate is large, sediment transport is strong with a large suspended load, and bedforms are in phase with the free surface.



[Figure 8.4 from Julien]

Here is a beautiful [video](#) on **antidunes** migrating upstream Medano Creek at Great Sand Dunes National Park & Preserve in Colorado.

c. Progression from 2D to 3D ripples and dunes

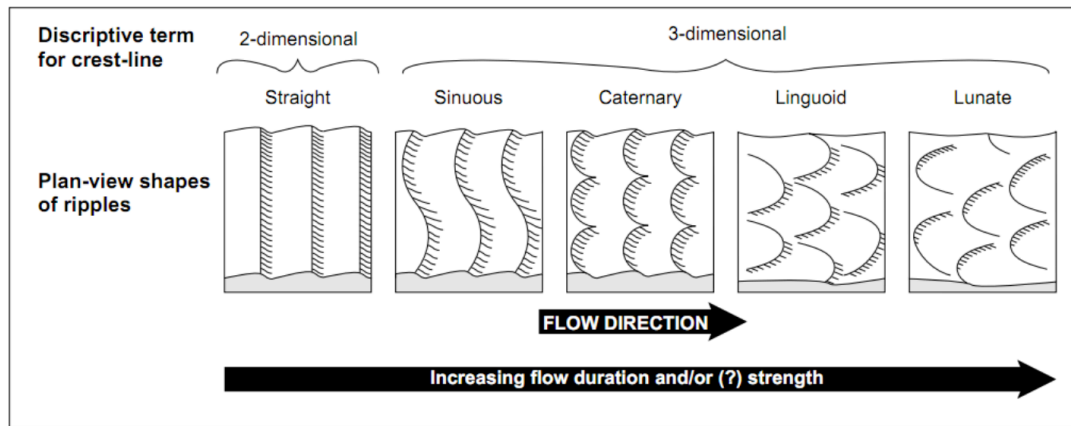


Figure 5-5. Terms used to describe plan forms of asymmetrical bedforms (ripples and dunes). After Allen, 1968 and Blatt, Middleton and Murray, 1980.

d. Active bedform migration infer background flow? In a new research study (Silvestro et al., 2020), scientists show for the first time that large sand ripples known as megaripples (dunes) are migrating in Martian deserts. Megaripples are distinguished from smaller ripples by the coarser sand grains that gather at their crests and their larger size: the ripples range from 30 centimeters (1 foot) to tens of meters across. The below video shows movement of Martian megaripples in several locations on the Red Planet. You can see the slight change in the ripples' position over the course of seven to ten years.

Until the publication of the above-mentioned research, scientists thought megaripples were relics of Mars's more geologically active past. Megaripples develop taller crests and migrate more slowly because their coarser grains are more difficult for wind to move.

Scientists thought it was impossible for the Red Planet's thin atmosphere—100 times less dense than Earth's—to conjure up powerful enough winds to move these coarser crests. But the new study shows wind can blow hard enough to move megaripples, suggesting the Martian atmosphere is more dynamic than previously thought.

1.2 2. Cross-shore sediment transport

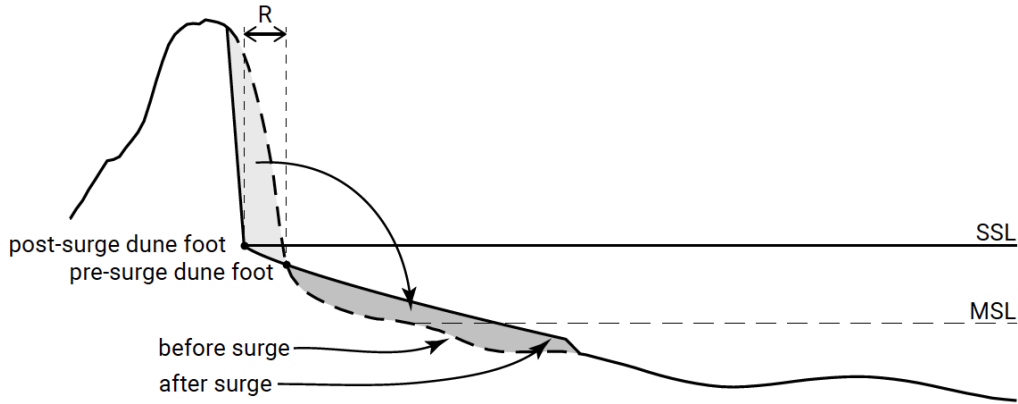


Figure 10.5: Dune erosion during a severe storm surge redistributes sediments in the cross-shore direction. There is no loss of sediment in the alongshore direction and hence the sediment will return under moderate conditions, through the combined action of small waves (under water) and onshore wind (on the beach).

1.3 3. Long-shore sediment transport

1.3.1 1). Erosion (coastline recession) vs deposition (coastline accretion)

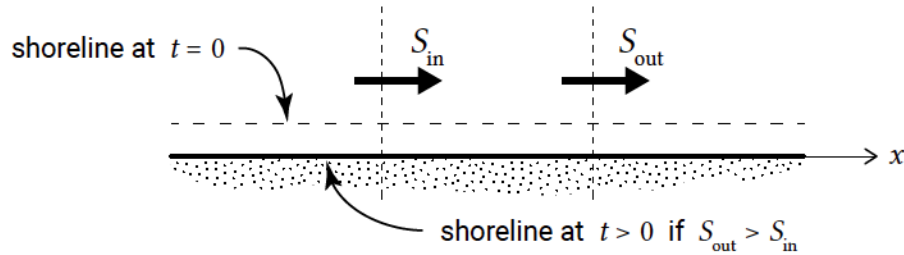


Figure 8.6: Longshore transport continuity.

The **long-term coastline changes** are governed by net, yearly-averaged longshore sediment transport rates, averaged over several wave conditions with varying magnitudes and directions. The very presence of sediment transport does not lead to either erosion or deposition.

- If one considers a portion of beach as sketched in plan in the above figure 8.6 (coastline is aligned with the x-axis), the coastline will remain stable as long as S_{in} is equal to S_{out} .
- On the other hand, if S_{out} is greater than S_{in} (S increases as we move along the beach, $\frac{dS}{dx} > 0$, or sediment transport divergence), material must be **eroded** in the area under consideration in order to maintain a sediment mass balance. The coastal profile, including the shoreline, will **recede** (see below Figure 8.8).
- The shoreline will move seawards (**accretion**) if the longshore sediment transport decreases in the transport direction ($\frac{dS}{dx} < 0$ or sediment transport convergence). A sediment mass balance relates morphological changes to sediment transport gradients, and to sediment sinks and sources.

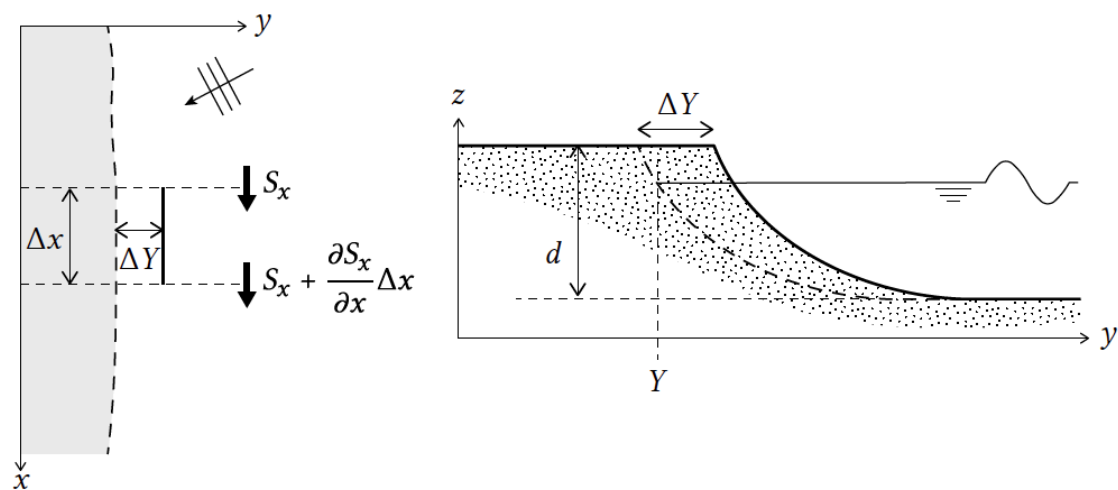


Figure 8.8: Longshore transport S_x and coastal change ΔY in a coastline model.

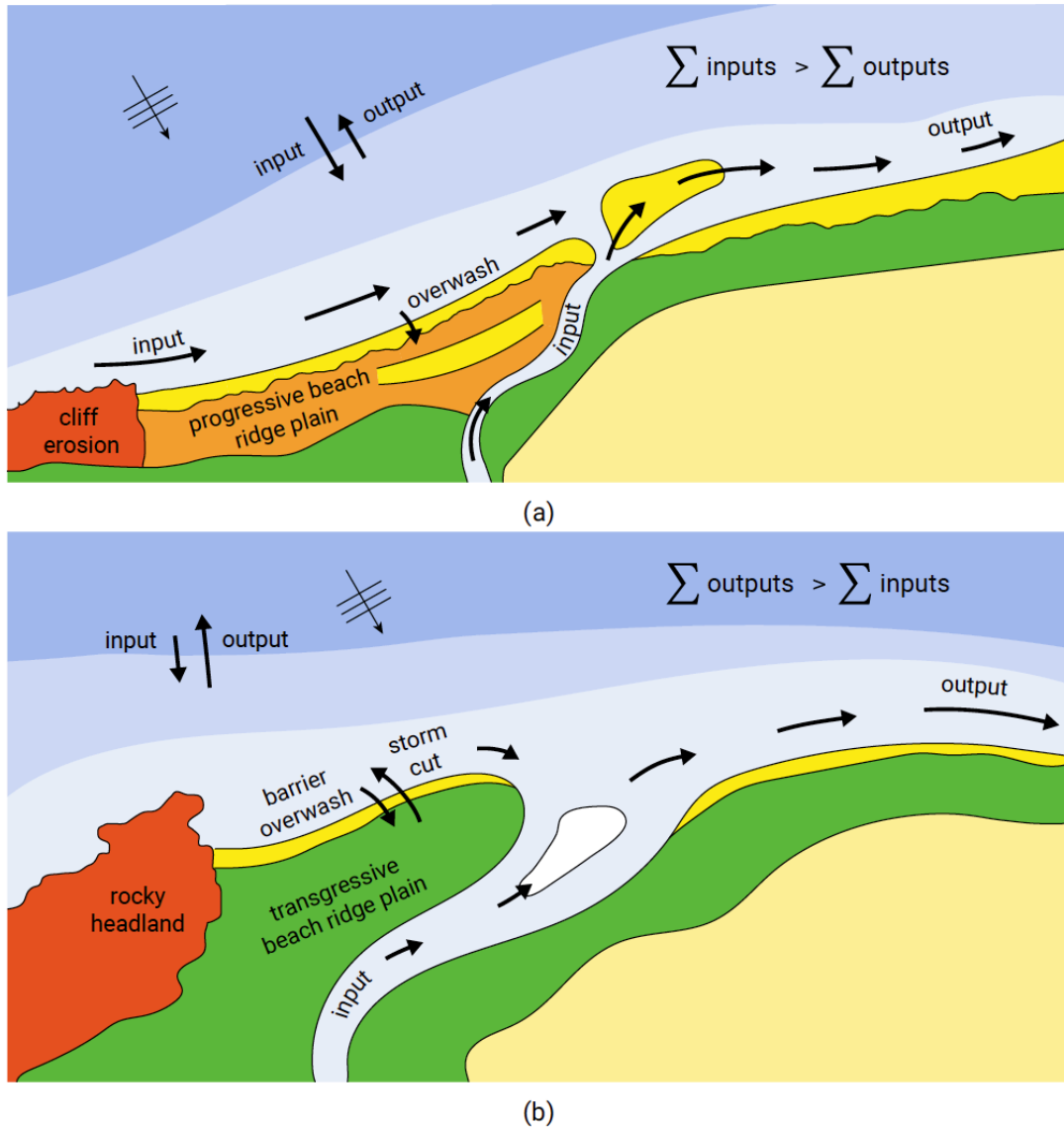


Figure 1.16: Sediment budgets along (a) deltaic and (b) estuarine coasts. In the example, the deltaic coast gains sediment; the sum of the inputs is larger than the sum of the outputs. This is reversed for the estuarine coast that loses sediment.

1.3.2 2). Coastal features and coastal change due to longshore transport

In this section, we consider the structural coastal response to large disturbances or interruptions (natural and human-induced) that can be explained based solely on longshore transport gradients. The onshore and offshore movements of sediment that take place on short timescales, such as storm-induced erosion and seasonal variations, are **assumed to cancel out on the longer timescales of shoreline change**. Any net gains or losses of sediment in the cross-shore direction (e.g. due to offshore canyons or inland aeolian transport) can be considered as sinks or sources.

Natural disturbances are for instance coastline interruptions by tidal basins or rivers, the presence of an offshore island and river sediment supply. Human-induced disturbances are for instance shore protection structures, river regulation works, nourishment schemes and maintenance dredging.

a. Blockage of longshore transport by shore-normal structures

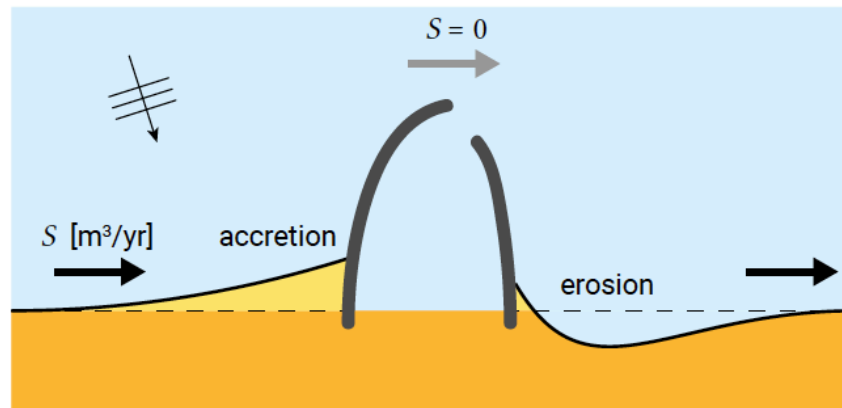


Figure 10.2: Typical structural downdrift erosion and updrift accretion.

b. Shadow effects due to obstacles away from the shoreline An obstacle in front of a coast, such as a rocky outcrop, an offshore breakwater or even a shipwreck, will reduce the wave activity in the zone of wave shadow between the object and the shore. Since the reduced wave activity in the shadow zone will result in a reduced sediment transport capacity, **material being carried along the shore will be deposited in the shadow zone**, forming a tombolo. Initially only a shoal will form, which we call a salient. This can, however, develop into a point of land connecting the original shoreline to the obstacle (see below Figs. 8.20 and 8.21). We now speak of tombolo. A tombolo completely blocks longshore transport in the zone landward of the obstacle.

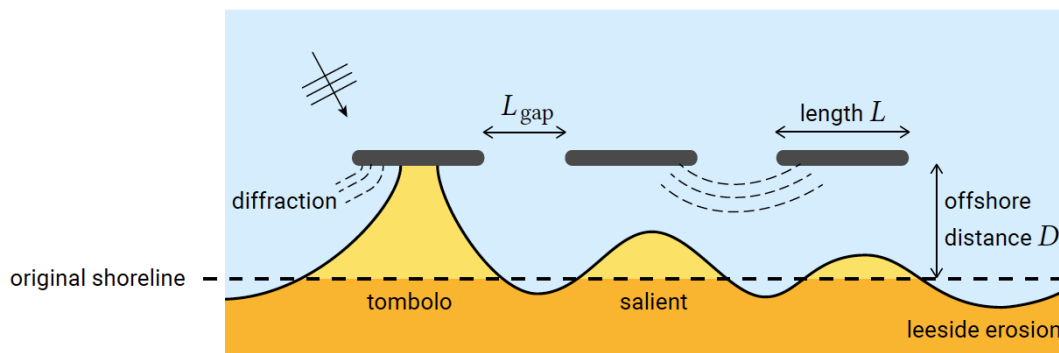


Figure 10.19: Detached shore-parallel breakwaters.



Figure 8.20: The holocene tombolo of Mariscal formed behind an island in Bombinhas, Brazil. The photo was taken from a hill in the original island (see black arrow in map) which is now connected to the Porto Belo peninsula by the tombolo. Photo by Carolina Piccoli, map Open-StreetMap ('Credits' on page 575).



Figure 8.21: Tombolo formed behind a detached breakwater in Alicante, Spain. Photo by Leonid Andronov ('Credits' on page 575).

c. Interrupted coasts: spits A spit is a pointed tongue extending into the sea. Spits develop where the longshore transport capacity is diminished due to coastline interruptions (river, estuary, end of island). The direction of a spit usually is a continuation of the shoreline from which sediment is supplied.

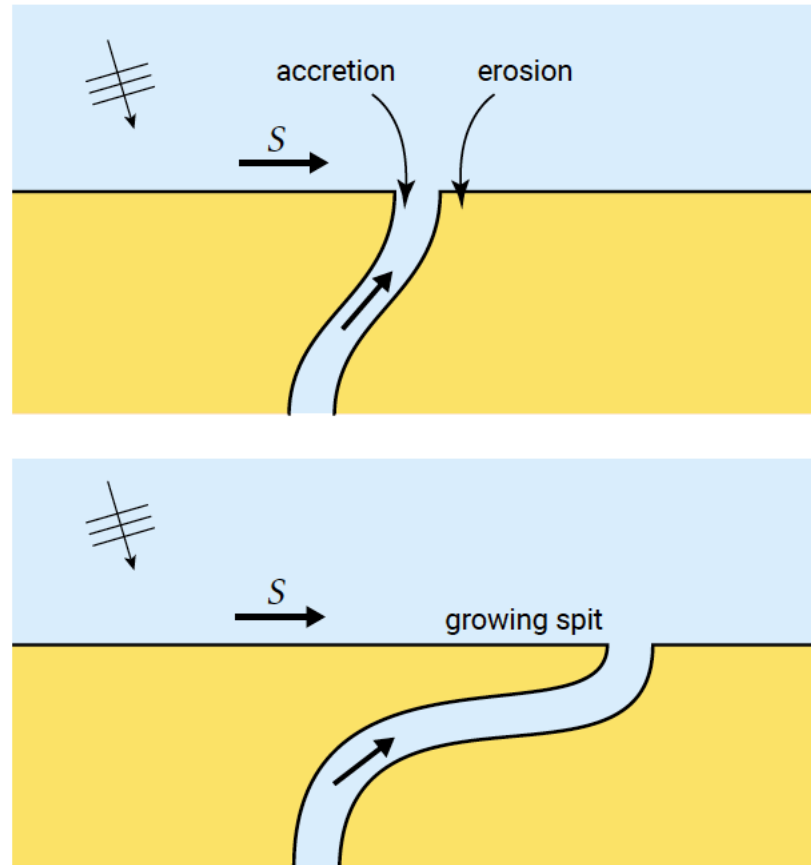


Figure 8.28: Spit and river mouth.

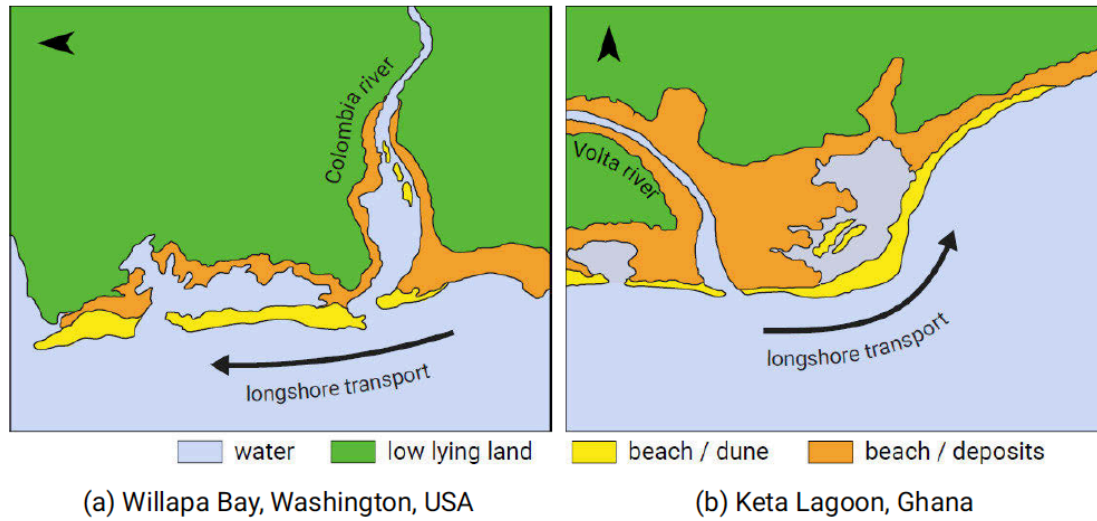


Figure 8.29: Barrier spit development as a result of longshore transport.

d. Deltaic coastlines Built primarily from river-borne sediments, deltas form when the amount of sediment delivered at the mouth of a river exceeds the amount removed by waves and tidal currents. The main effect of waves and tides is to redistribute the sediment along the coast.

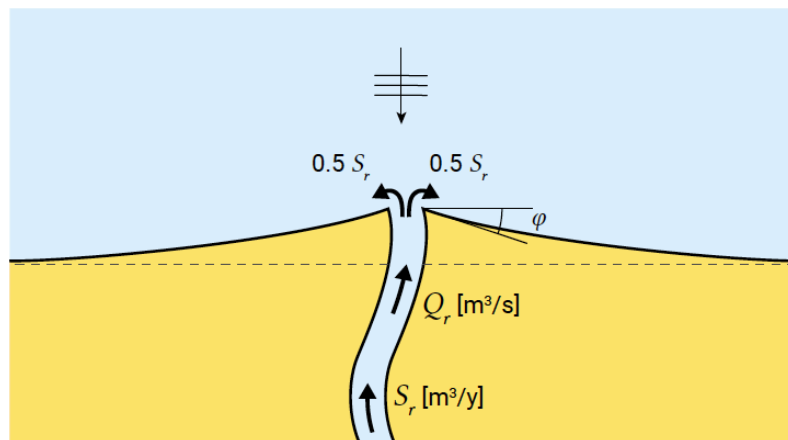


Figure 8.30: Development of deltaic coastline with waves that are normally incident to the original coastline and without tidal influence. The river discharge is Q_r and the sediment discharge S_r . The angle ϕ is determined by the sediment discharge and wave conditions.

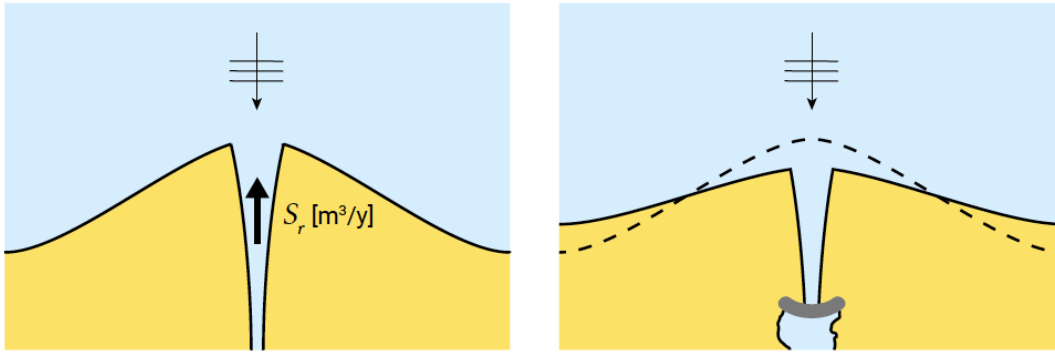


Figure 8.31: Delta erosion after the sediment supply is cut off. Due to wave-induced longshore transport gradients, sediments are redistributed along the coast, resulting in erosion of the delta (cf. the development of a shoreline perturbation in Sect. 8.4.4).

1.4 3. Coastal protection strategies and methods

1.4.1 1). Management strategies

In the last decade of the last century, especially in the context of raising awareness of the increased pressures on the coastal zone due to possibly accelerating climate change, management strategies to deal with coastal erosion were broadly divided into three categories: - **Retreat**: do nothing and accept retreat and possibly flooding (as happened during the successive deglaciation periods since the earth came into existence). - **Accommodate**: adapt coastal infrastructure to resist the increased risk of erosion and flooding. - **Protect**: take protective measures to counteract erosion and flooding.

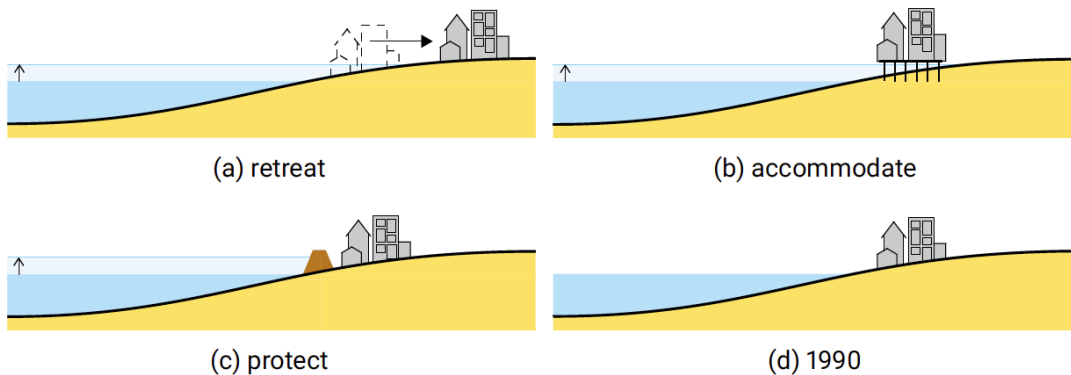
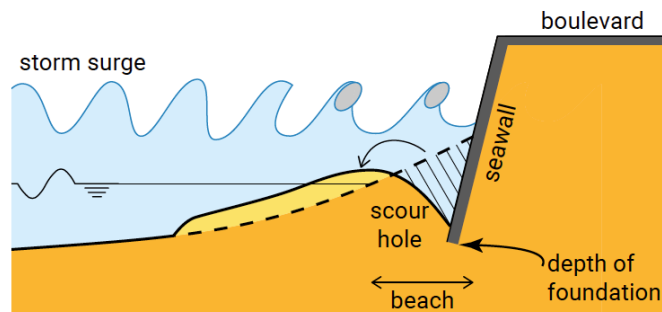


Figure 10.1: Management strategies for coastal erosion according to IPCC (1992) and World Coast Conference (1993): (a) retreat, (b) accommodate and (c) protect. Panel (d) refers to a specific situation in which a certain coastline (in this case the 1990 coastline in the Netherlands) is preserved using mainly sand nourishments.

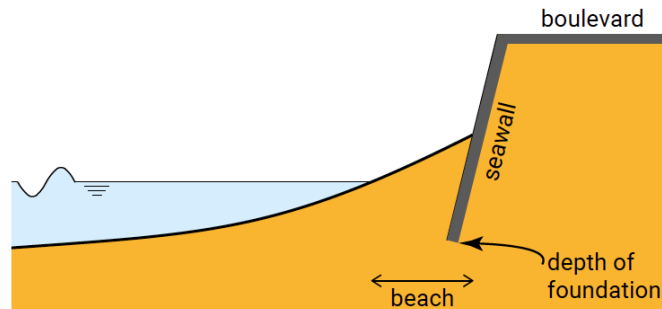
1.4.2 2). Selection of protection method

In general, there are two possible solutions to coastal morphological problems, viz. “hard” measures (coastal structures) and “soft” (natural) measures: - **“Soft” methods**: The principle of “soft” measures like beach or foreshore nourishment is to compensate for the eroded sand by nourishing sand, without great interference in the sediment transport pat-

terns. - One example of a “soft” solution is managed retreat in combination with protecting the retreated shoreline with a restored marsh. - Another example is to bring sand from the accreting updrift shoreline to the eroding downdrift shoreline (e.g., jetties) - Another example is to bypass sediment from the upstream side of a hydropower dam to downstream in order not to interrupt the sediment supply to a river delta. Without such a system, river deltas may retreat after the construction of the dam. - “Hard” methods: Examples of “hard” measures are series of groynes, series of offshore breakwaters, submerged breakwaters and revetments or seawalls. They ensure that sediment is not eroded, or to a lesser extent, by interfering in the sediment transports in alongshore and cross-shore directions of the coasts. - Seawalls or revetments built parallel to the shore do not solve structural erosion because they fail to address the underlying cause—longshore transport gradients. While they may initially prevent dune or land loss, beach and underwater profile erosion continues, eventually leading to beach disappearance and potential damage or failure of the structures themselves.



(a) scour in front of a seawall or revetment



(b) maintaining a beach requires e.g. regular sand nourishments

Figure 10.23: Seawall or revetment separating the beach from the mainland (a boulevard in this case).

The choice between a “hard” or a “soft” measure depends both on the characteristics of the problem concerned and on economic considerations. In Coastal Zone Management practice, the use of beach nourishment is becoming increasingly popular. **Many of the frequently occurring adverse side effects of “hard” measures can be avoided by using beach nourishments.** It is noted, though, that beach nourishment in coastal zones that have a rich nearshore ecological value (e.g. sea grass meadows) is often problematic because of environmental reasons.

1.4.3 3). A tale of an eroded beach due to an incorrectly applied revetment

Figure 10.21 shown below shows the cross-shore and longshore damage of an incorrectly applied revetment. The picture is looking towards the north; to the south of this location, a non-periodic, episodic high river discharge event 25 years earlier had served as a source of sediment for two decades, counteracting sediment losses from Cua Dai beach in the northward direction. Around the beginning of the 21st century, the provincial authorities allowed resorts to be developed to the south of this location. However, the absence of a new episodic source event eroded the beaches in front of the resorts and **the resorts started to act as revetments and/or groynes**. Hence, the coast north of the resorts started to erode, so that the beaches – important for tourism – were disappearing. Although this was a problem of alongshore transport gradients causing erosion, it was attempted to halt this erosion by geotextile bags, preventing the crossshore supply of sediment to resolve the longshore transport erosive gradients. As a result two types of damage occurred: 1) although the erosion was diminished by the geotextile bags, wave attack on the bags caused toe erosion and damage to the bags' stability and the beaches in front disappeared; 2) the geotextile bags were initially successful in preventing further feeding of sediment to the alongshore transports, but shifted the alongshore transport gradients further to the north, causing erosion.



Figure 10.21: Cross-shore and longshore damage to an incorrectly applied revetment, Cua Dai beach, Hoi An, Vietnam, around 2016. Photo by Marcel J.F. Stive ([‘Credits’](#) on page 575).

1.5 References:

Silvestro, S., Chojnacki, M., Vaz, D. A., Cardinale, M., Yizhaq, H., & Esposito, F. (2020). Megaripple migration on Mars. *Journal of Geophysical Research: Planets*, 125, e2020JE006446. <https://doi.org/10.1029/2020JE006446>

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